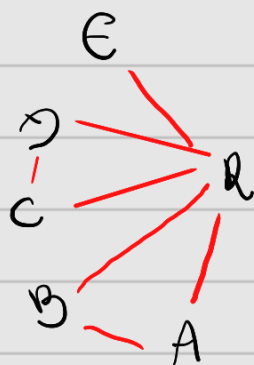


Exercice (MCM codigos)

1



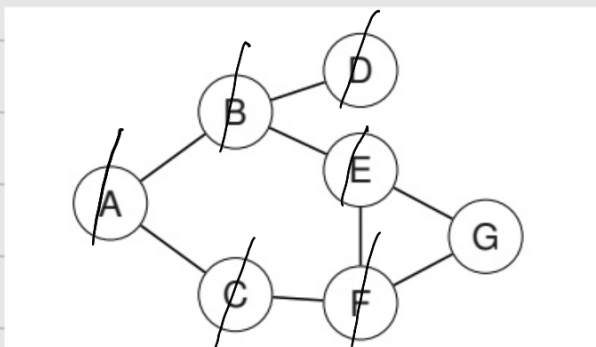
graus:

A-2 D-2
B-2 E-1
C-2 E-5

nao é possível, pois
para fazer busca
interiora, os graus
devem ser pares

2

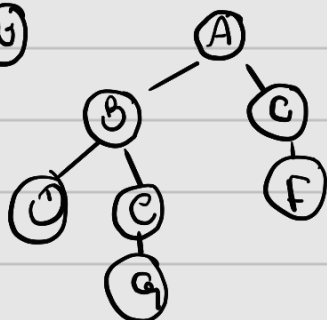
2



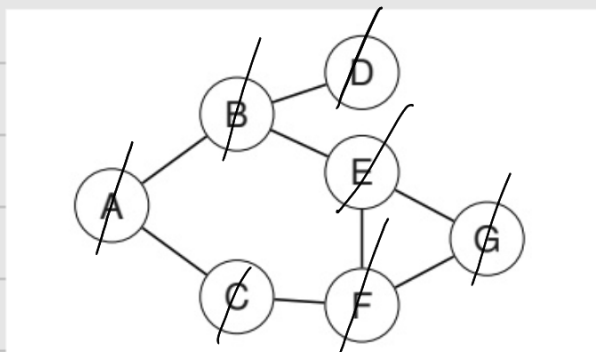
A-B-C-D-E-F-G

3

3 A-B-e-G

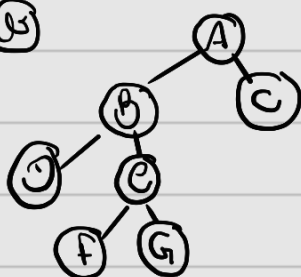


3 2



A-B-D-E-F-G-C

3



4) a) 6

* grafos ponderados deve seguir pelo menor peso na aresta

b) B - D - G - E

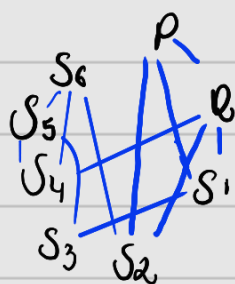
c)

Etapa	A	B	C	D	E	F	G	H
Início	$(\infty, -)$	$(0, -)$	$(\infty, -)$	$(\infty, -)$	$(\infty, -)$	$(\infty, -)$	$(\infty, -)$	$(\infty, -)$
1ª it (B)	$(1, B)$	$(0, -)$	$(\infty, -)$	$(2, B)$	$(\infty, -)$	$(\infty, -)$	$(4, B)$	$(6, B)$
2ª it (A)	$(1, B)$	$(0, -)$	$(9, A)$	$(2, B)$	$(\infty, -)$	$(\infty, -)$	$(4, B)$	$(5, A)$
3ª it (D)	$(1, B)$	$(0, -)$	$(9, A)$	$(2, B)$	$(\infty, -)$	$(4, D)$	$(3, D)$	$(5, A)$
4ª it (G)	$(1, B)$	$(0, -)$	$(9, A)$	$(2, B)$	$(6, G)$	$(4, D)$	$(3, D)$	$(5, A)$
5ª it (F)	$(1, B)$	$(0, -)$	$(5, F)$	$(2, B)$	$(6, G)$	$(4, D)$	$(3, D)$	$(5, A)$
Final	$(1, B)$	$(0, -)$	$(5, F)$	$(2, B)$	$(6, G)$	$(4, D)$	$(3, D)$	$(5, A)$

14) não direcionados: $\frac{n(n-1)}{2}$

direcionados: $n(n-1)$

15)



graus

P: 3

Q: 3

S6: 3

Q: 4

S4: 3

S3: 2

S1: 3

S5: 3

não é possível existir grafos eulerianos, pois há vértices ímpares.

16)

a) Grafo 1 não é direcionado, grafo 2 é direcionado

b) Grafo 1 não ponderado, Grafo 2 não ponderado

c) Grafo 1 simples, Grafo 2 simples

d) Grafo 1 conexo, Grafo 2 conexo

e) G1 não completo, G2 não completo

f) G1 não regular, G2 não regular

g) G1 possui ciclo: A-B-D-C-A

G2 possui ciclo: A-B-C-D-A

A B C D E F G

∞ ∞ ∞ 0 ∞ ∞ ∞

1^o (5, D) (9, D) ∞ 0 (5, D) (1, D) ∞

2^o (5, D) (9, D) ∞ 0 (4, F) (1, D) (2, F)

3^o (5, D) (9, D) ∞ 0 (3, G) (1, D) (2, F)

4^o (5, D) (4, G) (8, E) 0 (3, G) (1, D) (2, F)

17

a) $\{ \infty, \infty, \infty, 0, \infty, \infty, \infty \}$

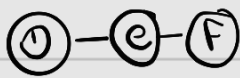
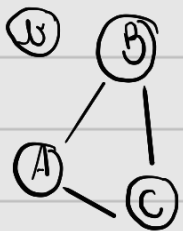
b) $\{ 5, 9, \infty, 0, 3, 1, \infty \}$

c) $\{ 5, 4, 8, 0, 3, 1, \infty \}$

d) 1^o: D, 2^o: F

18

a) grafo não direcionado, representado por matriz de adjacência



c) 2 componentes conexas

possui 1 ciclo: A-B-C-A

d) A: B, C

B: A, C

C: A, B

D: E

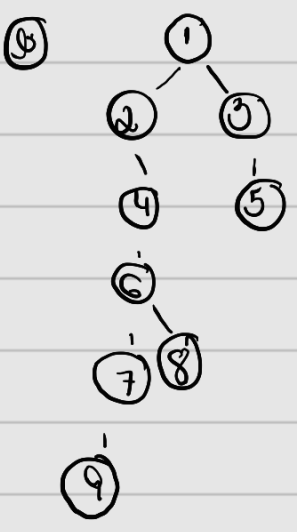
E: D, F

F: E

19) 9-2-1-4-7-12

20

a) 1-2-3-4-5-6-7-8-9

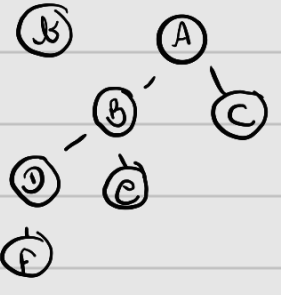


c) 1-2-4-6-7-9

d) 6 vertices

21

a) A-B-D-E-F-C



c) arestas que não pertencem à árvore de busca: F-D-E, e-n-c

d) o digrafo não possui ciclos direcionados

22

a) {0, ∞, ∞, ∞, ∞, ∞}

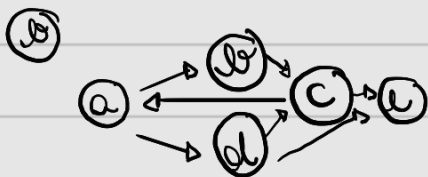
b) A-B-C-D-E-F

c) 6

d) A-B-C-E-F

	A	B	C	D	E	F
$ V $	0	∞	∞	∞	∞	∞
P_A	0	(2,A)	(5,A)	∞	∞	∞
$2^o B$	0	(2,A)	(3,B)	(5,B)	∞	(9,B)
$3^o C$	0	(2,A)	(3,B)	(5,B)	(5,C)	(9,B)
$4^o D$	0	(2,A)	(3,B)	(5,B)	(5,C)	(9,B)
$5^o E$	0	(2,A)	(3,B)	(5,B)	(5,C)	(6,E)
$6^o F$	0	(2,A)	(3,B)	(5,B)	(5,C)	(4,E)

23) a) grafo direcionado



c) grau de saída

A 2 C 2 E 0
B 1 D 2

d) grau de entrada

A 1 C 2 E 2
B 1 D 1

d) $A \rightarrow B \rightarrow E$

24)

a) nós

b) $S = \{A\}, \{B, C\}, \{D, E, F\}, \{G\}, \{H\}$

c) A, G, H

d) sim, $D - E - F - D$

e) graus:

A-0 D-2 G-0
B-1 E-2 H-0
C-1 F-2

25) a) M → S

b) não existe um ciclo envolvendo 4 nós de mais, existe menor, 4: $M \rightarrow S \rightarrow M$

c) graus 1 graus 2

in: 1

in: 2

out: 4

out: 0

total: 5

total: 2

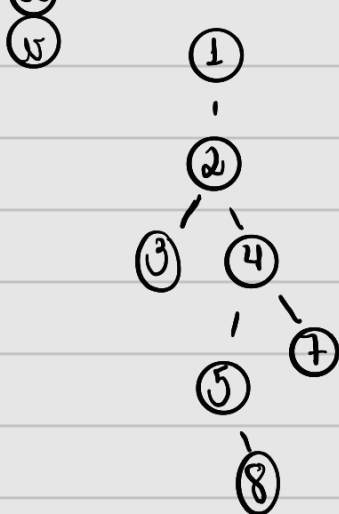
d) m: S, S
 n: D, S, Q, a
 O: m, m, n
 P: R, P, m
 Q: -
 R: -
 S: m, m

e)

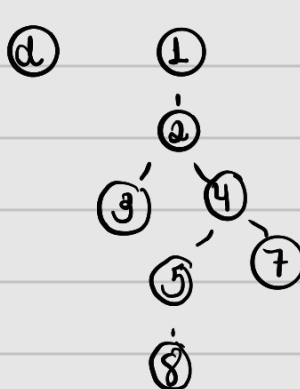
	m	n	o	p	q	r	s
m	0	0	0	0	0	0	2
n	0	0	1	0	1	1	1
o	2	1	0	0	0	0	0
p	1	0	0	1	0	1	0
q	0	0	0	0	0	0	0
r	0	0	0	0	0	0	0
s	2	0	0	0	0	0	0

26) graph 1:

a) 1-2-3-4-5-7-8

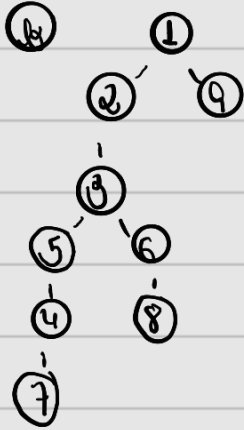


c) 1-2-4-7-5-8-3



graph 2:

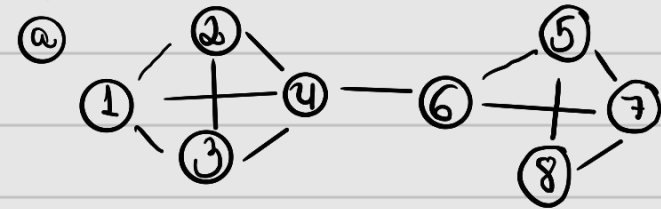
(a) 1-2-9-3-5-6-4-8-7



(c) 1-9-2-3-6-8-5-4-7



(27)



(28)

	1	2	3	4	5	6	7	8
1	0	1	1	1	0	0	0	0
2	1	0	1	1	0	0	0	0
3	1	1	0	1	0	0	0	0
4	1	1	1	0	0	1	0	0
5	0	0	0	0	0	1	1	1
6	0	0	0	1	1	0	1	0
7	0	0	0	0	1	1	0	1
8	0	0	0	0	1	0	1	0

- (c)
- 1: 2, 3, 4
 - 2: 1, 3, 4
 - 3: 1, 2, 4
 - 4: 1, 2, 3, 6
 - 5: 6, 7, 8
 - 6: 4, 5, 7
 - 7: 5, 6, 8
 - 8: 5, 7

28) BFS:

A - E - F - B - G - C - D - H

DFS:

A - B - F - C - D - H - G - E

distâncias até A, usando BFS:

$d(A): 0$ $d(E): 1$ $d(F): 1$ $d(B): 1$

$d(G): 2$ $d(C): 2$ $d(D): 3$ $d(H): 4$

29) a) 0 - 2 - 5 - 7 - 6 - 9 - 4 - 1



c) 0 - 2 - 7 - 5 - 6 - 9 - 4 - 1



30) início A

	A	B	C	D	E	F	G
início	0	∞	∞	∞	∞	∞	∞
1º A	0	(5, A)	(3, A)	∞	∞	∞	∞
2º C	0	(5, A)	(3, A)	(10, C)	(10, E)	∞	∞
3º B	0	(5, A)	(3, A)	(10, C)	(8, B)	∞	(6, B)
4º G	0	(5, A)	(3, A)	(10, C)	(7, G)	∞	(6, B)
5º E	0	(5, A)	(3, A)	(9, E)	(7, G)	(8, E)	(6, B)
6º D	0	(5, A)	(3, A)	(9, E)	(7, G)	(8, E)	(6, B)
7º F	0	(5, A)	(3, A)	(9, E)	(7, G)	(8, E)	(6, B)

a) 0, 5, 3, 9, 7, 8, 6

b) A, A, E, G, E, B

c) A: A

B: A $\rightarrow$ B

C: A $\rightarrow$ C

G: A $\rightarrow$ B $\rightarrow$ G

E: A $\rightarrow$ B $\rightarrow$ G $\rightarrow$ E

F: A $\rightarrow$ B $\rightarrow$ G $\rightarrow$ E $\rightarrow$ F

D: A $\rightarrow$ B $\rightarrow$ G $\rightarrow$ E $\rightarrow$ D

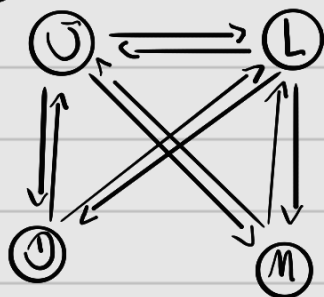
Para início em B o mesmo estilo

31

a) $V = \{ \text{Jensen, Leuzinger, Dufour, Medeiros} \}$

b) $J \rightarrow L, J \rightarrow O, J \rightarrow M$
 $L \rightarrow J, L \rightarrow O, L \rightarrow M$
 $O \rightarrow J, O \rightarrow L$
 $M \rightarrow J, M \rightarrow L$

c)



d) $J: L, O, M$ (abusive, adequate shower for complete)
 $L: J, O, M$
 $O: J, L$
 $M: J, L$

e)	J	L	O	M
J	0	1	1	1
L	1	0	1	1
O	1	1	0	0
M	1	1	0	0

(32) (a)	0	1	2	3	4	5
initials	∞	∞	∞	∞	∞	0
$1^{\circ} 5$	(2,5)	(4,5)	∞	(6,5)	∞	0
$2^{\circ} 0$	(2,5)	(4,5)	∞	(6,5)	(17,0)	0
$3^{\circ} 1$	(2,5)	(4,5)	∞	(6,5)	(17,0)	0
$4^{\circ} 3$	(2,5)	(4,5)	(18,3)	(6,5)	(17,0)	0
$5^{\circ} 4$	(2,5)	(4,5)	(18,3)	(6,5)	(17,0)	0
$6^{\circ} \infty$	(2,5)	(4,5)	(18,3)	(6,5)	(17,0)	0

(b) $5 \rightarrow 3-2$

(c) 18

(33)

(a)	1	2	3	4	5
1	0	0	0	1	2
2	0	1	1	1	1
3	0	1	0	1	0
4	1	1	1	0	0
5	2	1	0	0	0

1: 4, 5, 5
 2: 2, 3, 4, 5
 3: 2, 4
 4: 1, 2, 3
 5: 1, 1, 2

(b)	1	2	3	4	5	6	7
1	0	1	1	0	0	0	0
2	1	0	1	1	0	1	1
3	1	1	0	0	1	1	1
4	0	1	0	0	0	1	0
5	0	0	1	0	0	0	1
6	0	1	1	1	0	0	1
7	0	1	1	0	1	1	0

1: 2, 3
 2: 1, 3, 4, 6, 7
 3: 1, 2, 5, 6, 7
 4: 2, 6
 5: 3, 7
 6: 2, 3, 4, 7
 7: 2, 3, 5, 6

(c)	1	2	3	4	
1	0	3	1	0	1: 2, 2, 2, 3
2	0	1	1	1	2: 2, 3, 4
3	0	0	0	1	3: 4
4	0	0	0	0	4: -